

Github URL: <https://github.com/D6-prebuxofa/ASP.NET-WebAPI-Assignment-Arunkumar.git>

Q1)

Q1

GET /api/q1/welcome

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'https://localhost:44378/api/q1/welcome' \
-H 'accept: */*'

```

Request URL

```
https://localhost:44378/api/q1/welcome

```

Server response

Code

Details

200

Response body

```
Welcome to 5125!

```

Download

Response headers

```
content-encoding: gzip
content-type: text/plain; charset=utf-8
date: Wed, 05 Feb 2025 04:02:46 GMT
server: Microsoft-IIS/10.0
vary: Accept-Encoding
x-powered-by: ASP.NET

```

Responses

Code

Description

Links

200

OK

No links

```
C:\Users\auunku>curl -X GET "https://localhost:44378/api/q1/welcome"
Welcome to 5125!
C:\Users\auunku>
```

File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help Search WebApplication1

Q8Controller.cs Q7Controller.cs Q6Controller.cs Q5Controller.cs Q4Controller.cs Q3Controller.cs Q1Controller.cs

WebApplication1

namespace Assignment.Controllers

{

[ApiController]

[Route("api/q1")]

0 references

public class Q1Controller : ControllerBase

{

/// <summary>

/// Returns a welcome message.

/// </summary>

/// <returns>A welcome string message</returns>

[HttpGet("welcome")]

0 references

public IActionResult GetWelcome()

{

return Ok("Welcome to 5125!");

}

}

Solution Explorer

Search Solution Explorer (Ctrl+)

Solution 'WebApplication1' (1 of 1 project)

WebApplication1

Connected Services

Dependencies

Properties

Controllers

Q1Controller.cs

Q2Controller1.cs

Q2Controller.cs

Q4Controller.cs

Q5Controller.cs

Q6Controller.cs

Q7Controller.cs

Q8Controller.cs

WeatherForecastController.cs

WeatherForecastController

appsettings.json

Dockerfile

Program.cs

WeatherForecast.cs

WebApplication1.http

100 %

No issues found

Ln: 10 Ch: 39 SPC CRLF

Output

Show output from: Build

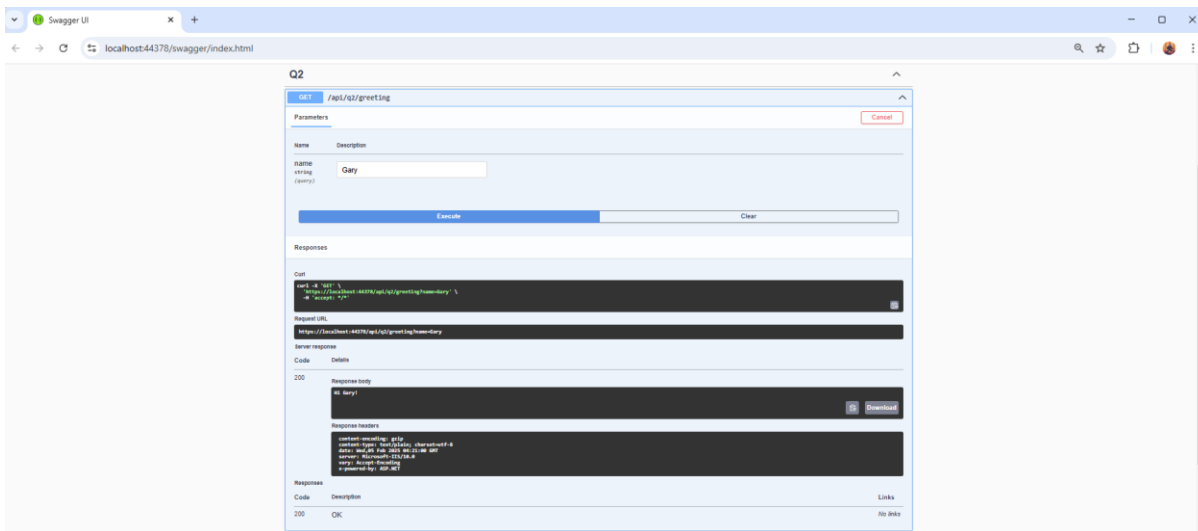
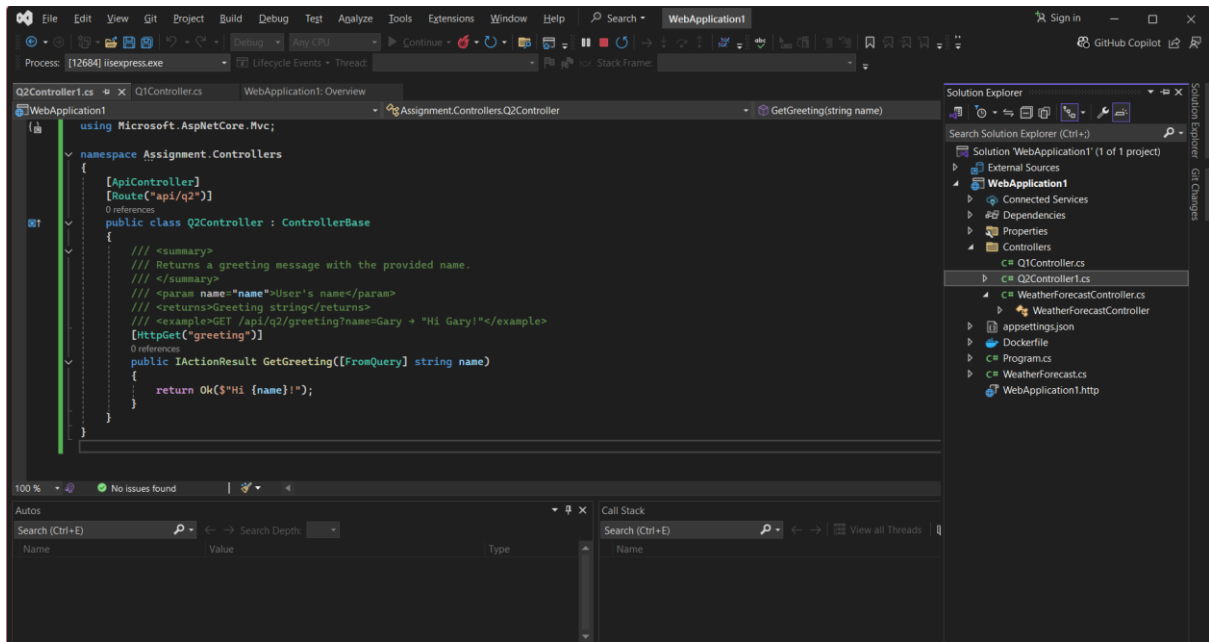
Build started at 23:55...

1----- Build started: Project: WebApplication1, Configuration: Debug Any CPU -----

===== Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped =====

===== Build completed at 23:55 and took 00.595 seconds =====

Q2)



```
C:\Users\Aunku>curl -k "https://localhost:44378/api/q2/greeting?name=Gary"
Hi Gary!
C:\Users\Aunku>
```

Q3)

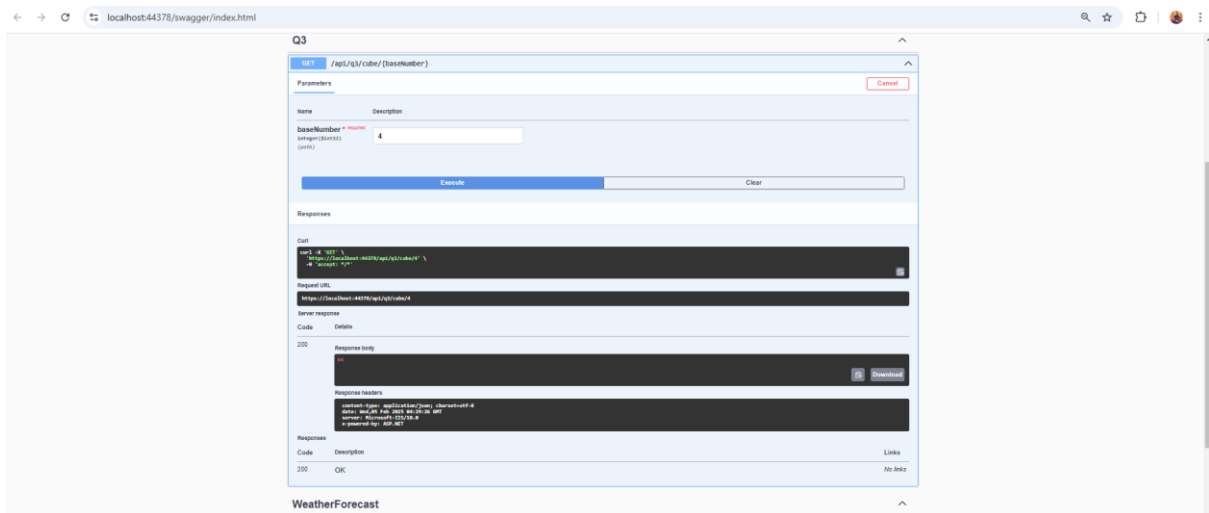
The screenshot shows the Visual Studio IDE with the 'WebApplication1' project open. The 'Q3Controller.cs' file is selected in the Solution Explorer. The code defines a controller for calculating the cube of a number. The Output window shows the build process, which completed successfully.

```
using Microsoft.AspNetCore.Mvc;

namespace Assignment.Controllers
{
    [ApiController]
    [Route("api/q3")]
    public class Q3Controller : ControllerBase
    {
        /// <summary>
        /// Returns the cube of the given integer.
        /// </summary>
        /// <param name="baseNumber">Base integer</param>
        /// <returns>Cubed value of the integer</returns>
        /// <example>GET /api/q3/cube/4 + 64</example>
        [HttpGet("cube/{baseNumber}")]
        public IActionResult GetCube(int baseNumber)
        {
            return Ok(Math.Pow(baseNumber, 3));
        }
    }
}
```

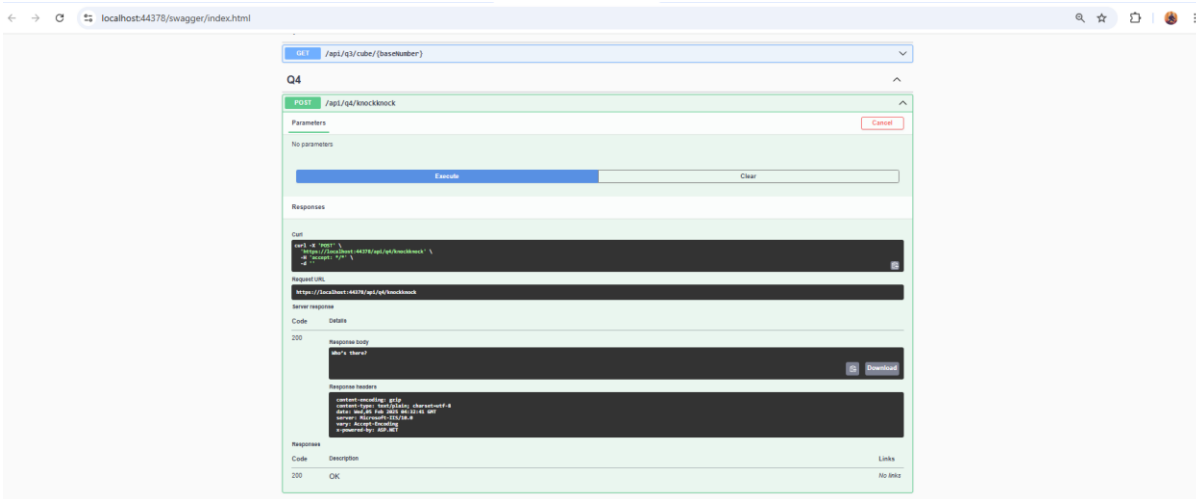
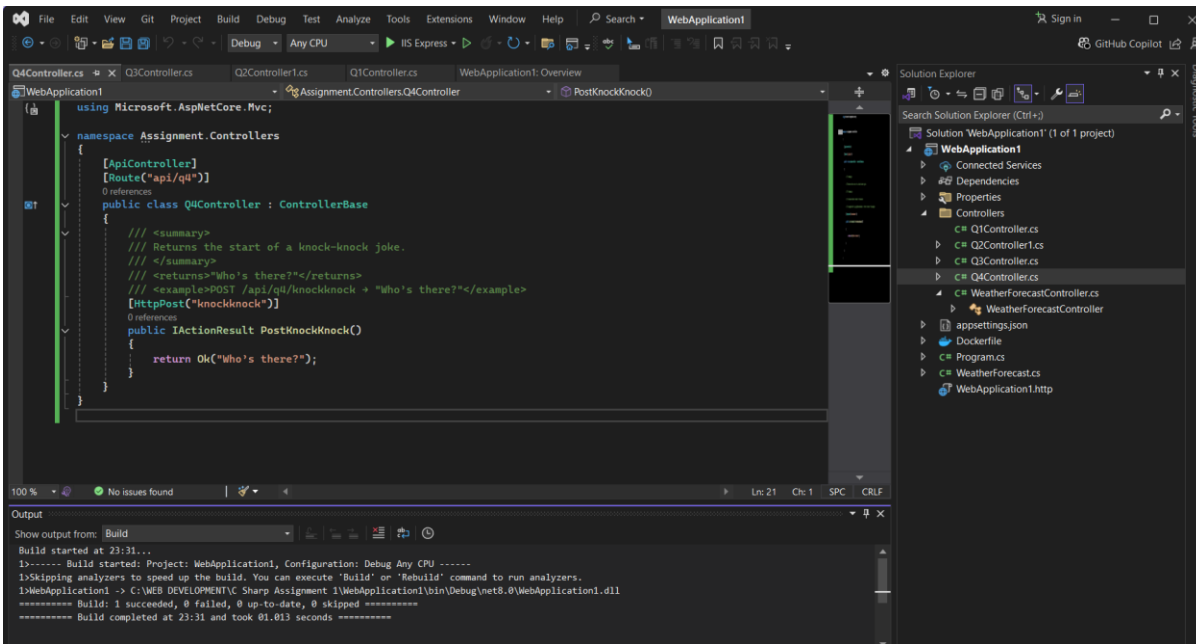
Output

```
Build started at 23:28...
1>----- Build started: Project: WebApplication1, Configuration: Debug Any CPU -----
1>Skipping analyzers to speed up the build. You can execute 'Build' or 'Rebuild' command to run analyzers.
1>WebApplication1 -> C:\WEB DEVELOPMENT\C Sharp Assignment 1\WebApplication1\bin\Debug\net8.0\WebApplication1.dll
***** Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped *****
***** Build completed at 23:28 and took 01.587 seconds *****
```



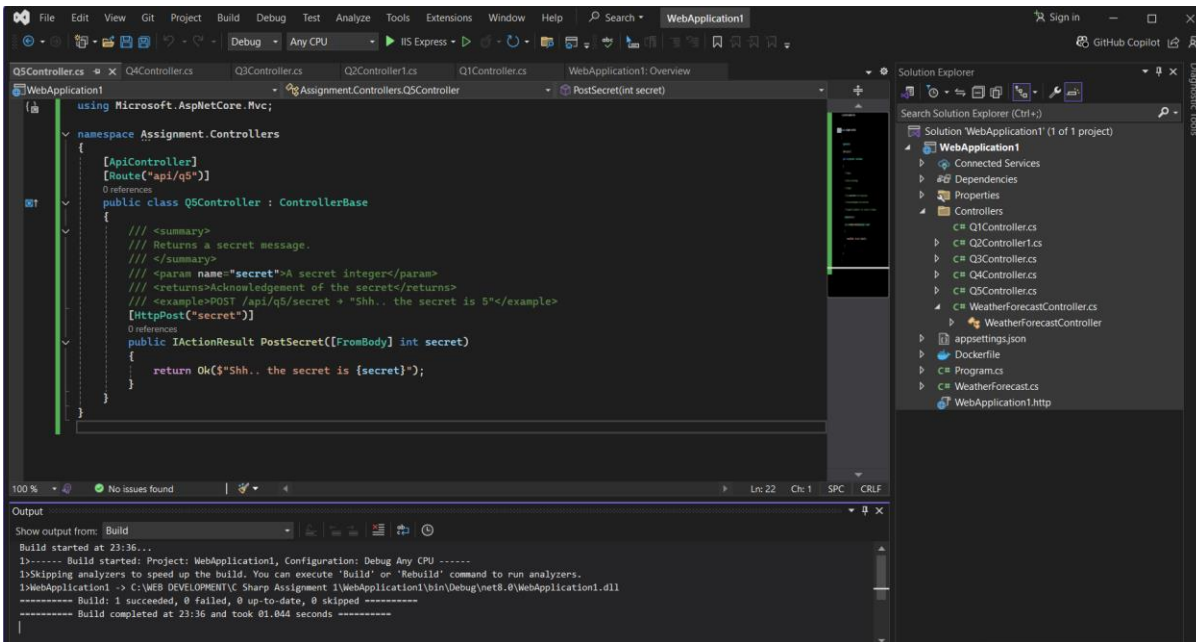
```
C:\Users\auunku>curl -k "https://localhost:44378/api/q3/cube/4"
64
C:\Users\auunku>
```

Q4)



```
C:\Users\anunku>curl -k -X POST -H "Content-Length: 0" "https://localhost:44378/api/q4/knockknock"
Who's there?
C:\Users\anunku>
```

Q5)

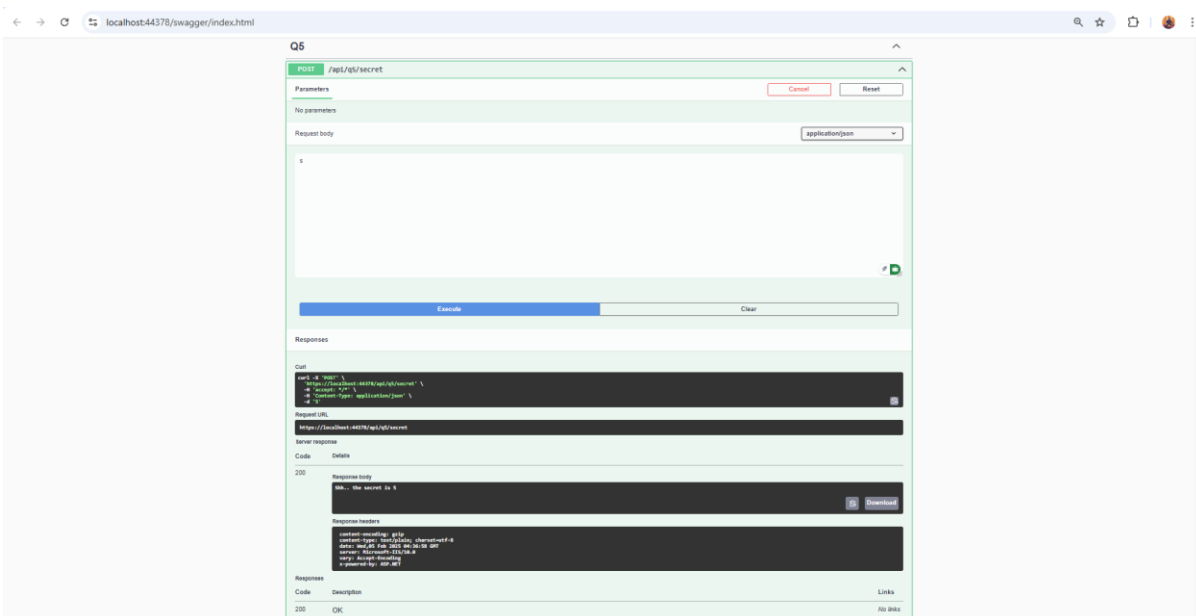


The screenshot shows the Visual Studio IDE with the following components:

- Code Editor:** Displays the `Q5Controller.cs` file. The code is as follows:

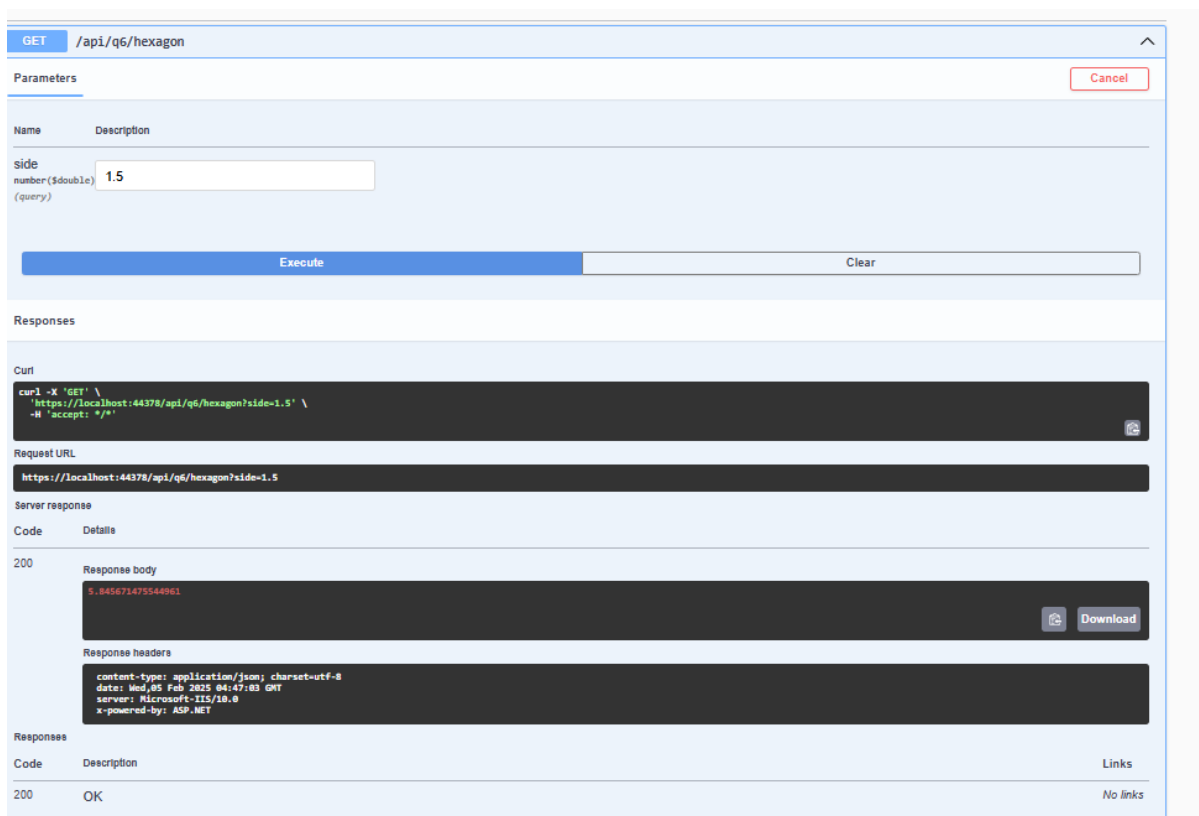
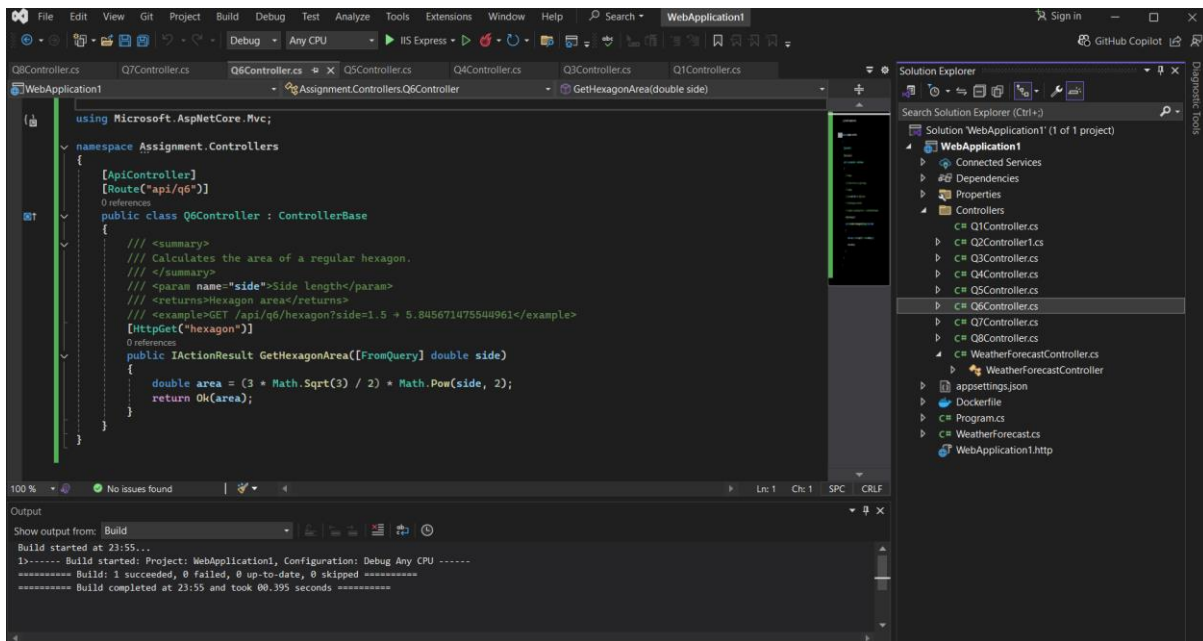
```
using Microsoft.AspNetCore.Mvc;

namespace Assignment.Controllers
{
    [ApiController]
    [Route("api/q5")]
    public class Q5Controller : ControllerBase
    {
        /// <summary>
        /// Returns a secret message.
        /// </summary>
        /// <param name="secret">A secret integer</param>
        /// <returns>Acknowledgement of the secret</returns>
        /// <example>POST /api/q5/secret + "Shh.. the secret is 5"</example>
        [HttpPost("secret")]
        public IActionResult PostSecret([FromBody] int secret)
        {
            return Ok($"Shh.. the secret is {secret}");
        }
    }
}
```
- Solution Explorer:** Shows the project structure for `WebApplication1`, including controllers like `Q1Controller.cs` through `Q5Controller.cs`, and other files like `appsettings.json`, `Dockerfile`, `Program.cs`, and `WeatherForecast.cs`.
- Output Window:** Shows the build output for the project. The output indicates that the build was successful and completed at 23:36, taking 01.044 seconds.



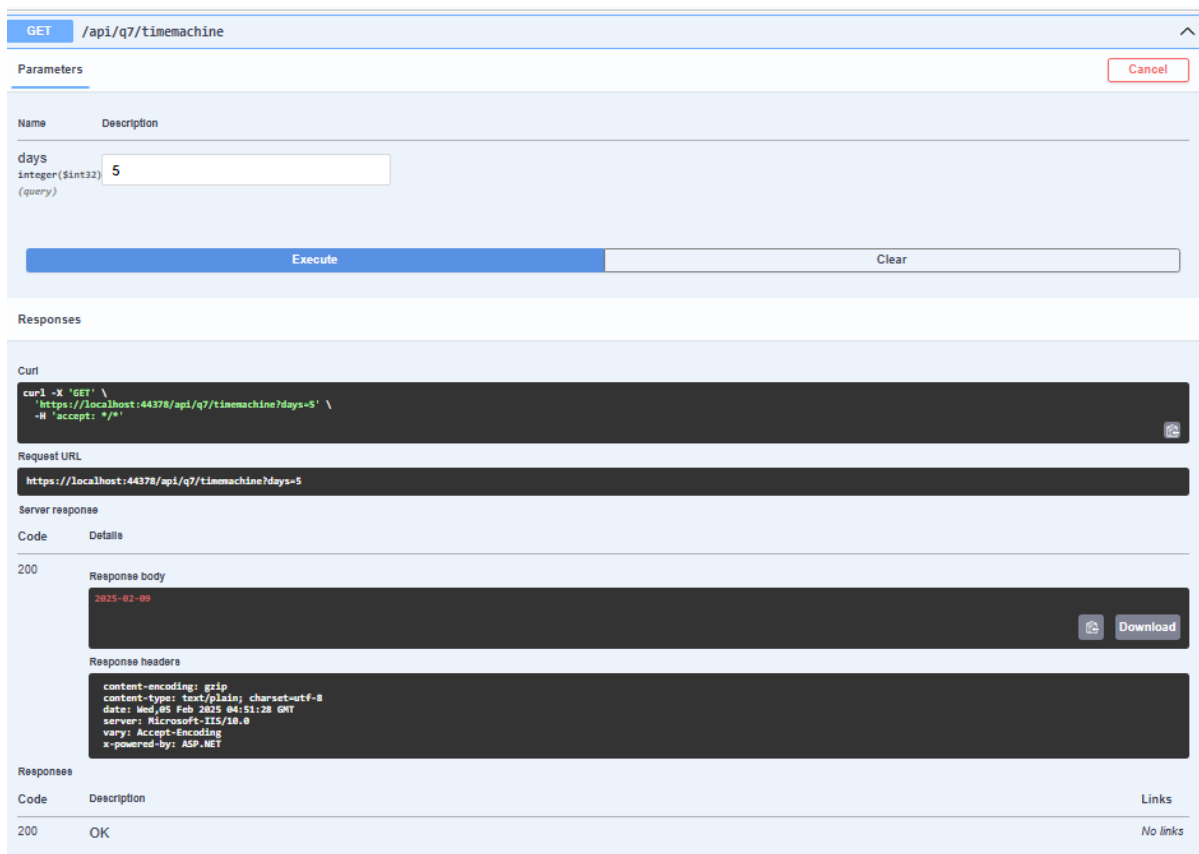
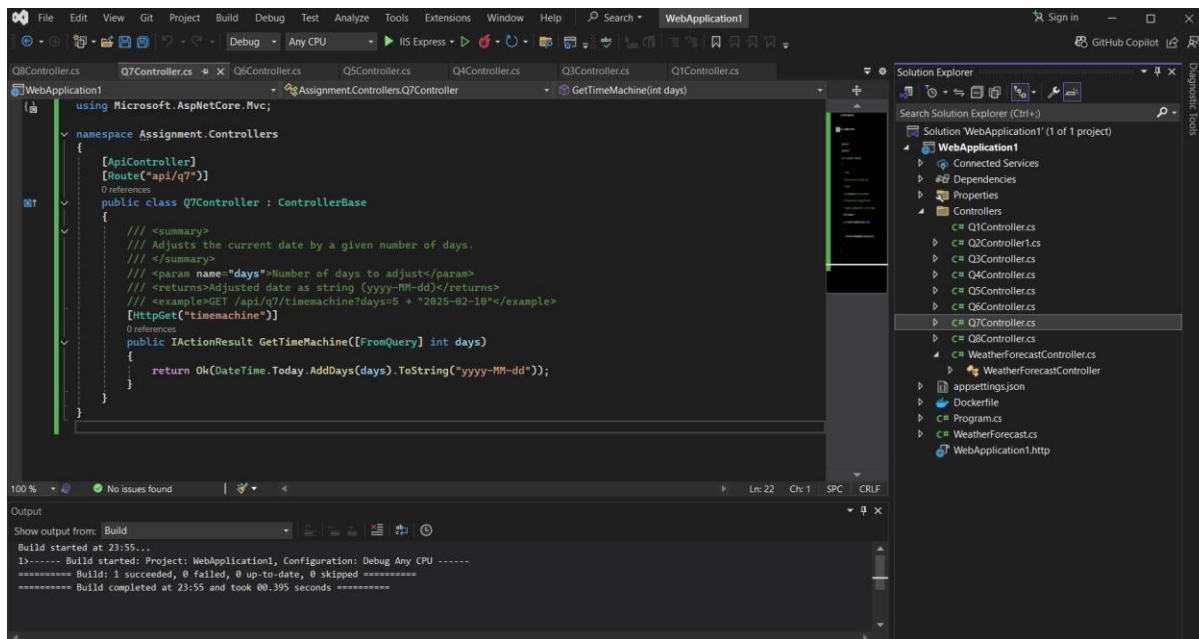
```
C:\Users\anku>curl -k -H "Content-Type: application/json" -d "5" "https://localhost:44378/api/q5/secret"
Shh.. the secret is 5
C:\Users\anku>
```

Q6)



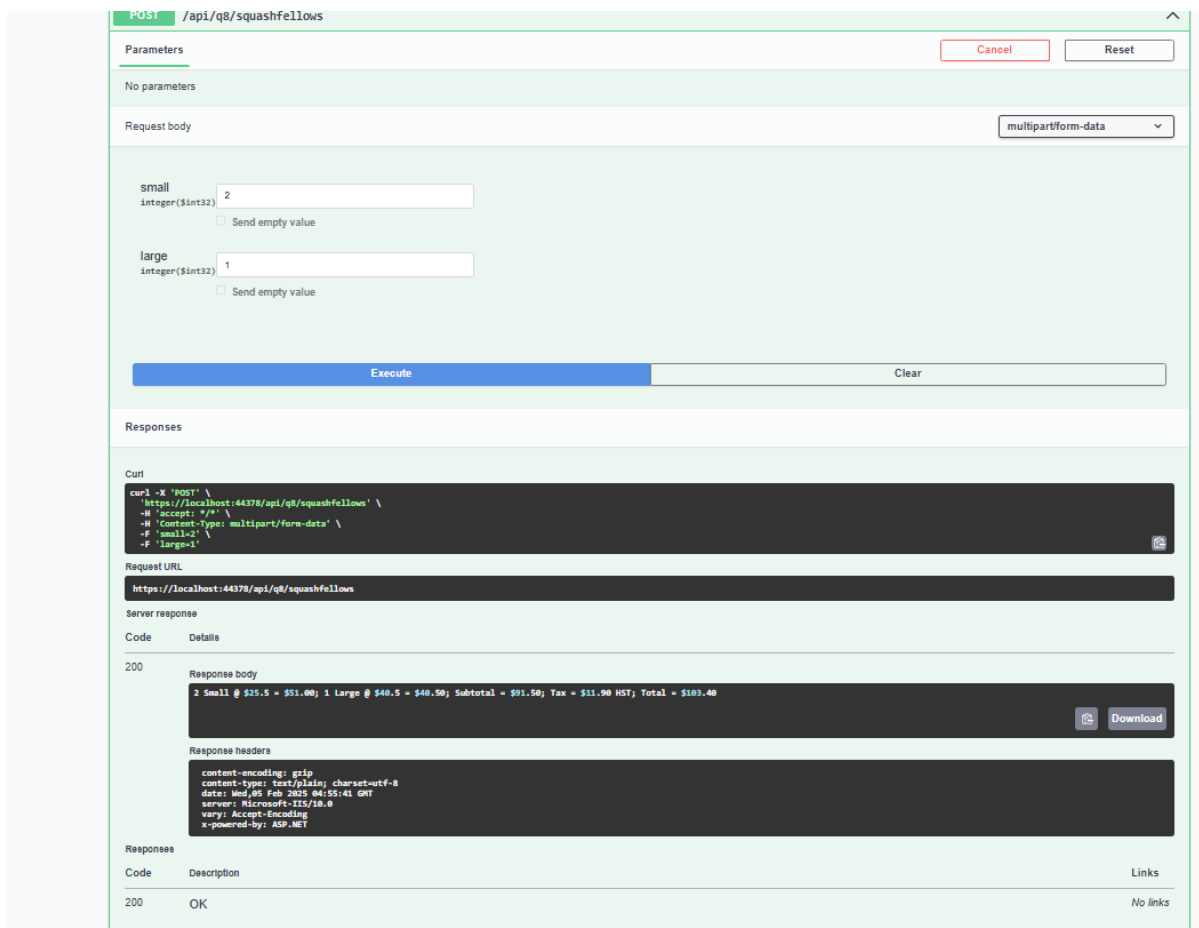
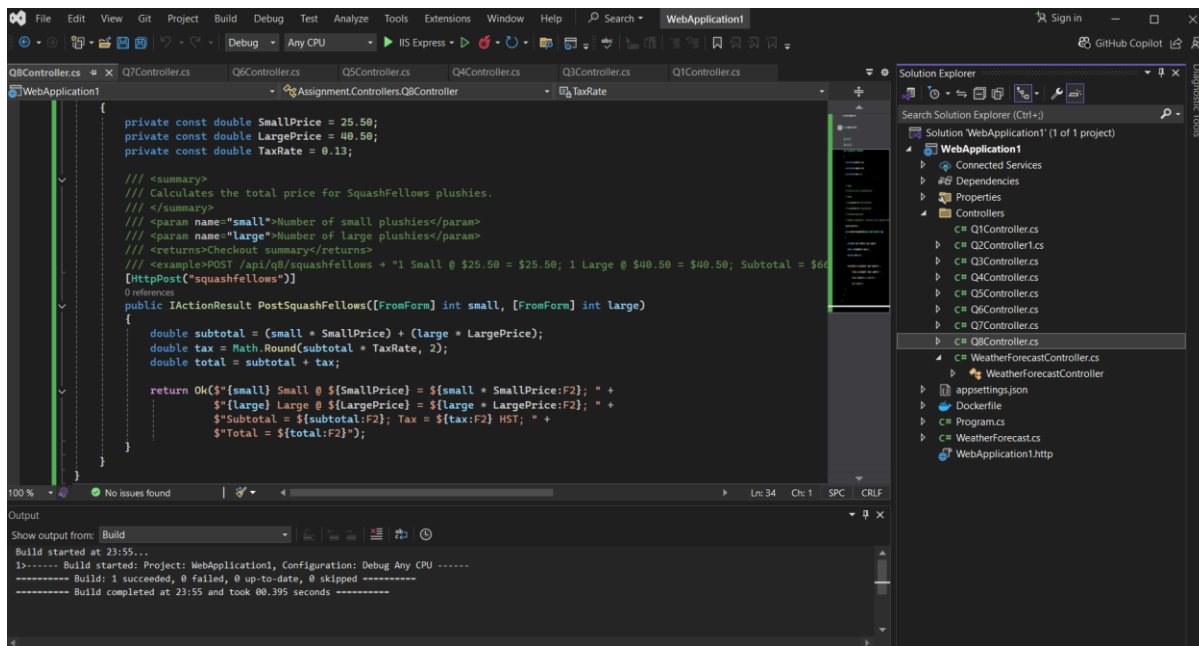
```
C:\Users\anunku>curl -k "https://localhost:44378/api/q6/hexagon?side=1.5"
5.845671475544961
C:\Users\anunku|
```

Q7)



```
C:\Users\aunku>curl -k "https://localhost:44378/api/q7/timemachine?days=5"
2025-02-09
C:\Users\aunku>
```

Q8)



```
C:\Users\auunku>curl -k -X POST -d "Small=2&Large=1" "https://localhost:44378/api/q8/squashfellows"
2 Small @ $25.5 = $51.00; 1 Large @ $40.5 = $40.50; Subtotal = $91.50; Tax = $11.90 HST; Total = $103.40
C:\Users\auunku>
```